

Week 6 Assignment: Linear Q-Function Approximation

Feature design, semi-gradient updates, and stability evidence

CMPSC 480 - Instructor Solution Deck

Instructor material - do not distribute with the question deck

Solution overview

This instructor deck provides a complete matched solution and grading guidance.

Coverage: Week 6: function approximation, semi-gradients, and the deadly triad

Points

10 total

Expected time

6-8 hours

Submission

Repository plus experiment memo

Instructor verification checklist

- ▶ Use Python 3.11 and NumPy; do not use a library that implements the assessed algorithm.
- ▶ Keep data, model, optimization, and evaluation responsibilities behind explicit interfaces.
- ▶ Validate shapes, ranges, finite values, empty inputs, and terminal or degenerate cases.
- ▶ Use deterministic seeds and record every configuration used for reported evidence.
- ▶ Include focused tests and a README with exact reproduction commands.

Solution 1.a - Features and dimensions (1 points)

Question focus. Define at least four state-action features and justify their scales.

Answer. A full-credit design names each signal, such as normalized coordinates, goal distance, wall risk, and action alignment; all are finite and on comparable documented scales.

Solution 1.b - Features and dimensions (1 points)

Question focus. Prove by an executable assertion that parameter size does not grow when the environment map grows.

Answer. Instantiate two map sizes with the same feature schema and assert equal feature and weight shapes. Reject any feature vector with a mismatched dimension.

Solution 2.a - Linear values and policy (1 points)

Question focus. Compute $Q\text{-hat}(s,a;w)$ and test a known vector by hand.

Answer. Return $\text{float}(w \text{ dot } f)$. For $w=[1,-2,0.5]$ and $f=[2,1,4]$, $Q\text{-hat}=2-2+2=2$.

Solution 2.b - Linear values and policy (1 points)

Question focus. Specify training and evaluation tie behavior.

Answer. Training explores uniformly with probability epsilon and otherwise uses a stable argmax over legal actions. Evaluation sets epsilon to zero and uses the same documented stable tie order.

Solution 3.a - Semi-gradient update (2 points)

Question focus. Derive the update from one-half times squared TD error.

Answer. With y treated as fixed, negative loss gradient is $(y - \hat{Q}) \text{grad } \hat{Q}$. For a linear function $\text{grad } \hat{Q} = f$, so w receives $\alpha \text{ times } \delta \text{ times } f$.

Solution 3.b - Semi-gradient update (1 points)

Question focus. Compute an update for $w=[0,0]$, $f=[1,2]$, $\text{reward}=1$, $\text{gamma}=0.5$, $\text{max-next-}Q=4$, and $\text{alpha}=0.1$.

Answer. The target is 3, current Q is 0, delta is 3, and the new weights are $[0.3,0.6]$.

Solution 3.c - Semi-gradient update (1 points)

Question focus. State and test the terminal target.

Answer. When done is true, y equals reward with no next-state maximum. A test uses an enormous next Q and confirms the terminal update is unchanged.

Solution 4.a - Experiment and stability (1 points)

Question focus. Run one feature ablation and compare with a tabular small-state oracle.

Answer. Hold seed, budget, and hyperparameters fixed; report return and action agreement. Explain whether the removed feature changes generalization or bias.

Solution 4.b - Experiment and stability (1 points)

Question focus. Define an instability gate using at least three logged quantities.

Answer. A suitable gate watches nonfinite values, weight norm, TD-error quantiles, and evaluation return across seeds, and stops or flags the run when declared thresholds are exceeded.

Edge-case reasoning and expected behavior

- ▶ The prediction is zero and no coordinate changes.
- ▶ Target equals reward only.
- ▶ Raise a clear error before the dot product.
- ▶ Use stable evaluation order.
- ▶ Stop, record the transition and configuration, and mark the run invalid.

Grading rubric

Category	Points	Full-credit evidence
Features and dimensions	2	Features are fixed length, normalized, documented, and safely validated.
Linear values and policy	2	Dot product, legal-action filtering, probabilities, and ties are correct.
Semi-gradient update	4	Target, TD error, gradient, terminal rule, and signs are correct.
Experiment and stability	2	Controlled comparison includes return, TD error, parameter norm, and multiple seeds.

Reflection exemplar

Answer. A strong response links feature overlap to which predictions changed, cites controlled evidence, and distinguishes improved return from stable value estimates.

Grading guidance. Award credit for a specific claim supported by technical evidence from the submitted work.